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Activation with Intrinsic-Extrinsic Consensus

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https://github.com/horrible-dong/AIEC

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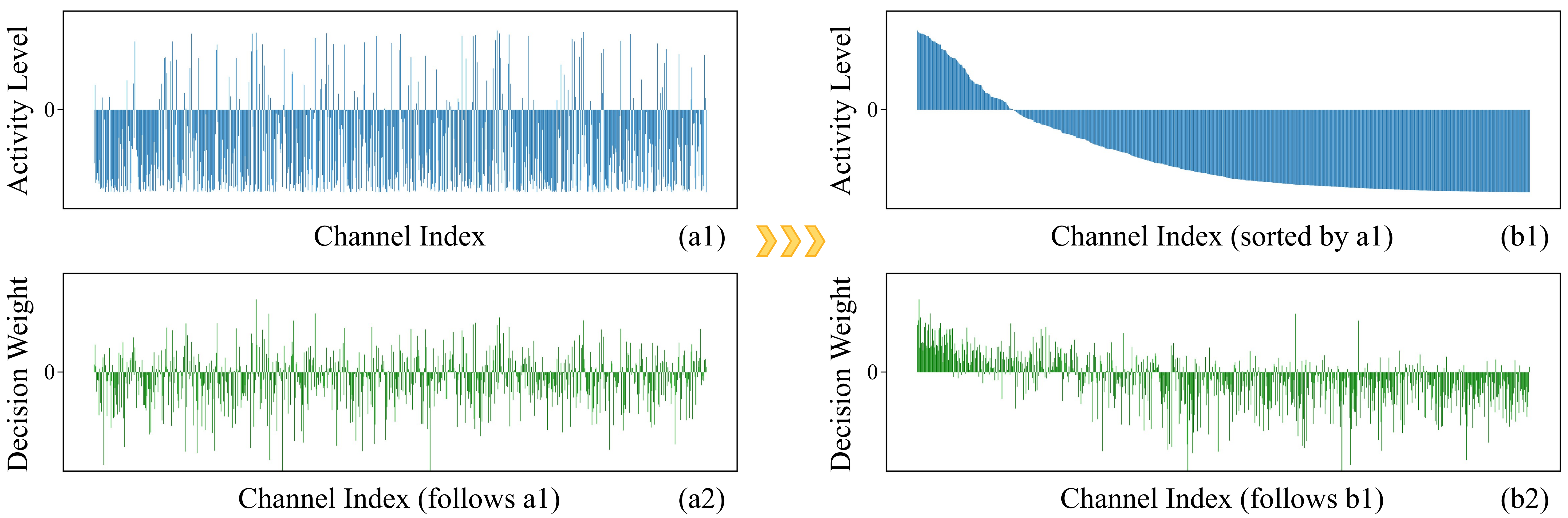
Highlights

- This work reveals that feature channel relevance can be jointly reflected by intrinsic activity levels and extrinsic decision weights, and observes a strong consensus between these two complementary assessments.
- A novel activation mechanism called Activation with Intrinsic-Extrinsic Consensus (AIEC) is proposed, which identifies irrelevant channels through the consensus between the intrinsic and extrinsic assessments, and applies channel-wise gatekeeping during training to suppress their activation responses while preserving informative channels.
- Extensive experiments demonstrate that AIEC effectively suppresses irrelevant channels and promotes sparser feature representations. Furthermore, AIEC is compatible with a wide range of mainstream ANN architectures and achieves superior performance compared to existing activation mechanisms across multiple tasks and domains.

Observations



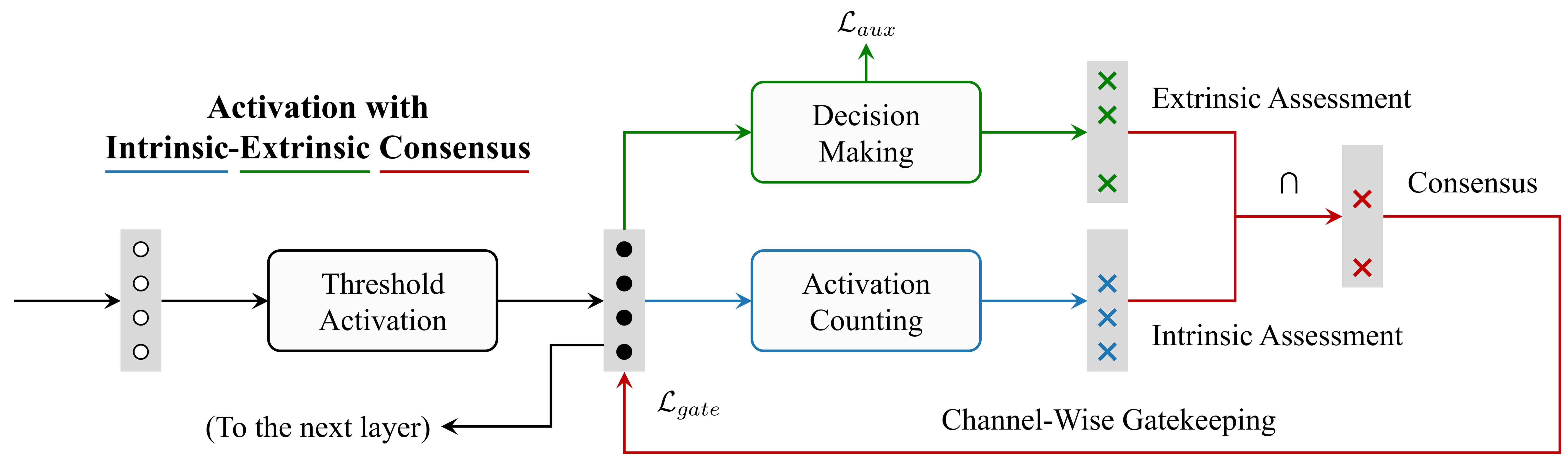
The intrinsic activity level (a1) and extrinsic decision weight (a2) are recorded for each channel of the feature vector after ReLU in the final block of ViT-Tiny, specifically for the "truck" category in the CIFAR-10 dataset. The activity level of a channel is computed based on its historical activation statistics over samples from the "truck" category, defined as "(number of activations - number of inhibitions) / total sample count". The decision weight is learned as the weight of a linear classifier applied to the post-activation feature vector. (b1) is (a1) sorted in descending order of activity level, and (b2) is (a2) re-indexed following the index order of (b1).

Observation 1: Variance in channel activity levels intrinsically reflects channel relevance.

Observation 2: Linear classifier acting on activated features learns distinct decision weights for each channel, extrinsically indicating channel relevance.

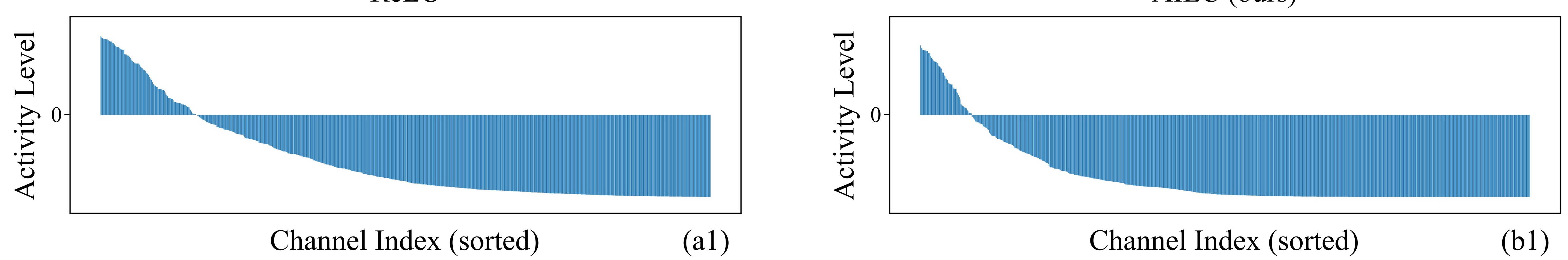
Observation 3: Intrinsic and extrinsic aspects show consensus for channel relevance assessment.

Method



Experiments

Channel activity levels under ReLU and the proposed AIEC, with AIEC yielding sparser representations.



Top-1 Acc / %		Softplus	ELU	SELU	SiLU	ReLU	GELU	GDN	AIEC
CIFAR-10	ViT-Tiny	84.3	82.0	79.4	85.5	89.9	89.2	81.8	**91.8**
	CaiT-XXS	82.5	80.7	78.4	86.6	89.4	88.7	80.0	**90.5**
	PVT-Tiny	90.6	89.3	85.4	92.5	93.0	92.8	82.8	**94.0**
	TNT-Small	88.3	85.4	83.7	90.5	90.8	91.1	85.1	**92.7**
CIFAR-100	ViT-Tiny	62.4	60.0	57.5	65.5	65.7	65.4	59.4	**71.1**
	CaiT-XXS	60.4	59.3	55.8	63.9	65.8	65.5	56.2	**70.0**
	PVT-Tiny	69.5	69.3	65.7	70.2	70.9	70.6	64.4	**76.0**
	TNT-Small	65.2	63.8	60.9	65.1	65.4	64.4	62.5	**73.9**
ImageNet-100	ViT-Tiny	74.1	68.9	66.4	74.1	75.4	76.4	67.9	**82.1**
	CaiT-XXS	70.9	69.1	65.9	76.1	76.0	76.7	69.5	**81.3**
	PVT-Tiny	79.5	77.1	76.1	79.5	81.9	81.4	75.8	**86.3**
	TNT-Small	78.9	79.3	76.4	77.6	79.9	77.2	76.9	**86.5**
ImageNet-1K	ViT-Tiny	70.0	64.2	63.1	66.9	70.9	70.4	65.2	**73.0**
	CaiT-XXS	70.3	68.1	66.7	73.2	74.0	73.6	66.1	**74.1**
	PVT-Tiny	71.5	69.2	68.5	72.8	73.7	73.5	66.5	**74.6**
	TNT-Small	72.0	70.7	70.3	71.5	73.4	73.3	68.2	**78.1**

Top-1 Acc / %		Softplus	ELU	SELU	SiLU	ReLU	GELU	GDN	AIEC
CIFAR-10	AlexNet	85.6	86.1	85.7	86.0	86.0	85.8	85.4	**86.3**
	VGG-11	91.3	92.0	91.5	91.9	**92.2**	91.9	91.1	**92.2**
	MobileNet	87.4	87.7	87.2	87.8	87.4	87.4	87.0	**89.0**
	ResNet-18	94.6	94.7	94.6	95.1	95.0	94.9	94.0	**95.1**
CIFAR-100	AlexNet	57.6	58.4	58.1	58.1	57.2	57.4	56.8	**58.9**
	VGG-11	69.6	69.9	69.7	69.9	70.2	70.0	70.1	**71.2**
	MobileNet	65.4	65.5	65.6	65.2	66.0	65.4	64.8	**66.1**
	ResNet-18	75.5	75.7	75.6	76.1	75.7	75.6	74.3	**77.0**
ImageNet-100	AlexNet	75.7	76.0	75.7	76.6	76.3	76.3	75.5	**79.2**
	VGG-11	87.0	87.3	87.6	87.8	87.7	87.5	86.7	**88.1**
	MobileNet	80.6	79.3	79.2	80.1	80.6	80.5	78.7	**80.8**
	ResNet-18	84.6	84.4	84.1	84.9	84.9	84.7	83.5	**86.5**
ImageNet-1K	AlexNet	56.1	56.3	56.1	56.4	56.5	56.4	55.6	**57.8**
	VGG-11	68.4	68.2	67.8	69.0	69.0	69.1	68.1	**70.1**
	MobileNet	67.2	66.7	67.1	67.4	68.1	68.2	66.3	**68.6**
	ResNet-18	69.3	69.4	68.9	69.7	69.7	69.4	68.3	**70.4**